The Sharp Krasnosel'skiĭ–Mann Constant Was Determined

Run fa_banach_001, agent_lane_16

2026-08-13

Source question

Let C be a bounded convex subset of a normed space, let $T : C \to C$ be nonexpansive, and let

$$x_n = (1 - \alpha_n)x_{n-1} + \alpha_n T x_{n-1}, \qquad 0 \le \alpha_n \le 1.$$

Cominetti, Soto, and Vaisman, arXiv:1206.4195, prove the universal bound

$$\|x_n - Tx_n\| \le \frac{1}{\sqrt{\pi}} \frac{\operatorname{diam}(C)}{\sqrt{\sum_{i=1}^n \alpha_i(1-\alpha_i)}}. \tag{1}$$

In their conclusion (source PDF page 12), they ask for the smallest universal constant κ that can replace $1/\sqrt{\pi}$. Their right-shift example only establishes $0.4688\ldots \le \kappa \le 1/\sqrt{\pi} = 0.5641\ldots$.

Later full answer

Bravo and Cominetti, arXiv:1606.05300, Theorem 1.1 (supporting PDF page 4), prove that the upper constant in (1) is tight. Precisely, for every $\kappa < 1/\sqrt{\pi}$ they construct a nonexpansive map

$$T : [0,1]^{\mathbb{N}} \longrightarrow [0,1]^{\mathbb{N}} \quad \text{in } \ell_\infty(\mathbb{N}),$$

an initial point, and a constant relaxation sequence $\alpha_n \equiv \alpha$ for which, at some iterate,

$$\|x_n - Tx_n\| > \kappa \frac{\operatorname{diam}(C)}{\sqrt{\sum_{i=1}^n \alpha_i(1-\alpha_i)}}.$$

Consequently no smaller universal constant works, while arXiv:1206.4195 supplies the matching upper bound. Hence

$$\boxed{\kappa_{\text{opt}} = \frac{1}{\sqrt{\pi}}}.$$

The construction is not merely an improvement of the source's affine right-shift example. The answering paper first formulates sharp recursive distance bounds as nested optimal-transport problems. Its Theorem 2.1 builds a nonlinear nonexpansive map on the infinite cube that attains all of those bounds simultaneously. A Markov-chain interpretation then gives the $1/\sqrt{\pi}$ asymptotics used in Theorem 1.1. This fully resolves the source question.

References

- R. Cominetti, J. A. Soto, and J. Vaisman, *On the rate of convergence of Krasnoselski–Mann iterations and their connection with sums of Bernoullis*, Israel J. Math. 199 (2014), 757–772; arXiv:1206.4195.

- M. Bravo and R. Cominetti, *Sharp convergence rates for averaged nonexpansive maps*, J. Math. Anal. Appl. 469 (2019), 60–81; arXiv:1606.05300.